MUGIC setup guide

Written October 2026. Attached: `mugic_capture.py`, a test script.

The MUGIC is a wearable motion sensor that streams its readings over WiFi to your computer, about 50 times a second. This guide gets you from a charged sensor to live numbers on screen in about 15 minutes.

What you need

- The MUGIC and its USB-C cable.
- A Mac or PC on the same WiFi network as the sensor.
- A network that asks only for a name and password: a home router or a phone hotspot. Networks that ask for a username (eduroam, most university and office WiFi) will not work.
- For the test script: Python 3.10 or newer and uv, which installs the script's one library for you.

1. Charge and power on

1. Charge with the USB-C cable. A full charge takes about 2 hours.
2. Hold the power button for more than 2 seconds to turn it on.
3. Watch the third LED. Solid blue means it is ready.

Third LED	Meaning
Solid blue	Own-network mode, ready to connect
Blinking blue	Own-network mode, sending data
Fast-blinking purple	Trying to join your WiFi
Slow-blinking purple	Joined your WiFi, sending data

The RESET button returns every setting to the factory state. Use it if you lose track of how the sensor is set up.

2. Connect it to your network

Out of the box, the sensor makes its own WiFi network, `MugicConnectXXXXXX` (the X's are unique to each sensor). Joining that network works, but your computer loses its internet connection while it's on it. **Client mode** fixes this: the sensor joins your WiFi instead, and both devices stay online.

You set up client mode once:

1. On your computer, join the sensor's network, `MugicConnectXXXXXX`. On our sensor, its password was `mugic` plus the same six characters as the network name.
2. Open `http://192.168.4.1/configure` in a browser.

3. Under **WiFi Client Mode Configuration**, enter your network's name and password, leave the port at `4000`, and press that section's **Send**.
4. Under **Wi-Fi IP List**, enter your computer's IP address on that network as **IP 1**, and press that section's **Send**.
5. Hold the **M** button for 2 seconds. The third LED blinks purple fast while the sensor tries to join your network, then slowly once it has joined.
6. Put your computer back on your normal network.

Things that commonly go wrong:

- **The name must match exactly.** An iPhone called "Sam's iPhone" actually uses a curly apostrophe (’). On a Mac, type it with Option + Shift +]. This cost us an afternoon.
- **2.4 GHz only.** The sensor can't see 5 GHz networks. On an iPhone hotspot, turn on **Maximize Compatibility** and keep the Personal Hotspot screen open while the sensor joins.
- **Passwords** can be up to 63 characters long.
- **"Configuration accepted" means saved, not connected.** If the LED goes back to solid blue, the sensor didn't manage to join.

Phone hotspot tip. An iPhone hotspot gives out addresses from `172.20.10.2` to `172.20.10.14`. Fill IP 1 to IP 8 with `172.20.10.2` through `172.20.10.9`, and the sensor reaches your computer whichever address it gets. On a home router, give your computer a fixed IP instead (a "DHCP reservation" in the router's settings).

To go back to the sensor's own network, hold **M** for 2 seconds again.

3. Check that data arrives

Put the attached `mugic_capture.py` in a folder and, in a terminal there, run:

```
uv run mugic_capture.py              # show live readings
uv run mugic_capture.py take1.csv    # show them and save every reading to a file
```

Press Ctrl-C to stop. Working output updates twice a second and looks like this:

```
  50 Hz | angles   126.5   -26.8    21.2 | gyro    18.9    31.1    -5.7 | battery 82%
```

- **"Hz"** should read about 50 at the default setting. The angles and gyro numbers change as you move the sensor.
- **Stuck on "Waiting for data"?** The sensor isn't reaching your computer. See Troubleshooting below.
- **"Address already in use"?** Another program is listening on port 4000, often the MUGIC plugin open in a music app. Close it and try again.

4. Settings worth knowing

The configuration page lives at `http://192.168.4.1/configure` when you're on the sensor's own network. In client mode, MUGIC's documentation says the same page is at the sensor's own address on your network. Each section has its own **Send** button, and **Reload with latest parameters** shows what's currently saved.

Setting	Default	What it does
Data Output Frequency (ms)	`20`	Time between readings. `20` is 50 per second; `10` is 100 per second, which we believe is the sensor's maximum.
Acceleration Source	Linear (gravity removed)	Acceleration with gravity subtracted. Good for detecting hits and stomps.
Data Message Prefix	`mugicdata`	The name each reading arrives under. The script expects `/mugicdata`.
Data Output Format	OSC	Leave as is. The test script reads OSC.
Data Output Mode	Wi-Fi	How data leaves the sensor. A USB mode also exists; this guide covers WiFi only.
Wi-Fi IP List (IP 1 to 8)	empty	Computers that receive the data in client mode.
Client and AP UDP Port	`4000`	The port data arrives on. Change it only if another program needs 4000.
AP SSID / AP PWD	`MugicConnectXXXXXX` / unique	The name and password of the sensor's own network.
BLE Name	per sensor	The sensor's Bluetooth name.
LED Mode	Normal	LED behaviour.

5. The data

Each reading is one OSC message, sent over UDP to port 4000 under the address `/mugicdata`. It holds 24 numbers in this order. MUGIC's documentation lists 23; the firmware we tested sends an extra battery-voltage value after the battery percentage.

Fields	What they are	Units and limits we saw
`AX AY AZ`	Acceleration	m/s² (gravity removed by default). Clips around ±40 per axis on hard, fast hits.
`EX EY EZ`	Orientation as Euler angles	Degrees
`GX GY GZ`	Rotation speed (gyroscope)	Degrees per second. Clips around ±2000 per axis.
`MX MY MZ`	Magnetometer	Raw sensor units
`QW QX QY QZ`	Orientation as a quaternion	Unitless

Fields	What they are	Units and limits we saw
`Battery`	Charge	Percent
`BattV`	Battery voltage (not in the docs)	Volts, about 4.1 when full
`SysStatus GyroStatus AccelStatus MagStatus`	Sensor calibration status	Small whole numbers
`Ms`	The sensor's clock	Milliseconds
`SeqNum`	Reading counter	Goes up by 1 per reading; a gap means a lost reading

The sensor chip is a Bosch BNO055. Its datasheet covers the measurement details. The units above come from our own recordings, not from MUGIC's documentation.

Troubleshooting

Symptom	Likely cause	Fix
LED goes back to solid blue after trying to join	Couldn't join your network	Check the network name character by character (curly apostrophes), the password, and that the network is 2.4 GHz
The `MugicConnectXXXXXX` network reappears	Same: the join failed	As above
LED blinks purple slowly, but the script shows nothing	Data goes to the wrong address, or your computer left the network	Check your computer's IP is in the IP list (`ipconfig getifaddr en0` on a Mac). Turn off auto-join for other networks so the computer stays put.
Script says "Address already in use"	Another program holds port 4000	Close the MUGIC plugin or any other OSC app
Still nothing	Firewall blocking incoming UDP	Allow incoming connections for Python, or briefly turn the firewall off to test
Settings are a mess	—	Press RESET to return to factory settings

More documentation

- MUGIC Getting Started: charging, buttons, first connection.
- MUGIC Troubleshooting: WiFi and LED problems.
- MUGIC Data API: the data message.
- MUGIC plugin guide for macOS: the official plugin for music software.
- MUGIC website
- Switching to WiFi client mode: a user's write-up.